

Higher Tier only questions

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks available |     |     |       |       |      |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3        | (a) | (i)  |  | <b>A, D and E</b> – all <b>three</b> needed (1)<br><br>they have equal numbers of protons and electrons (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 | 2   |     | 2     |       |      |
|          |     | (ii) |  | <b>B</b> and <b>F</b> – both needed for (1)<br><br><b>B</b> has a +/ +1/1+ charge and <b>F</b> has a +2/2+<br>– both needed for (1)<br><br>award (2) for <b>B</b> <sup>+</sup> and <b>F</b> <sup>2+</sup>                                                                                                                                                                                                                                                                                                                                                                                        |                 | 2   |     | 2     |       |      |
|          | (b) |      |  | any of following for (1) <ul style="list-style-type: none"><li>atoms having the same number of protons but different number of neutrons</li><li>same atomic number but different mass number</li><li>atoms of the same element having different number of neutrons / different mass number</li></ul><br>award (1) for comparison of <sup>12</sup> C and <sup>14</sup> C e.g. <ul style="list-style-type: none"><li><sup>12</sup>C has 6 protons and 6 neutrons <b>and</b> <sup>14</sup>C has 6 protons and 8 neutrons</li><li><sup>14</sup>C has two more neutrons than <sup>12</sup>C</li></ul> | 2               |     |     | 2     |       |      |
|          |     |      |  | Question 3 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2               | 4   | 0   | 6     | 0     | 0    |